Cost Effective Housing Techniques

Unit IV: Rural Housing

Syllabus

Rural housing: introduction - traditional practice of rural housing - mud housing technology - mud roofs - characteristics of mud - fire treatment for thatch roof - soil stabilization - rural housing programme.

1. Introduction

Nearly **two-thirds of Indians live in about 6.5 lakh villages**. A rural house is **home, workplace (farm, craft, animals), store and social space** together. Rural housing is built by the family itself, with **local materials** (mud, thatch, stone, bamboo, timber) and **traditional skills**. The main need is to **improve** these houses - for durability, safety and hygiene - **without losing the cost advantage**.

Features of rural housing

- **Self-help construction** with village masons, using **local materials**, little cash
- **Climate responsive** design: thick mud walls and thatch keep the house cool in summer and warm in winter
- House combined with **cattle shed, grain storage, courtyard (angan), work area**
- Built **incrementally** (room by room as money allows)
- **Problems:** poor durability, **damage from rain, flood, fire, wind and earthquake**, termite attack, smoke from cooking, no toilets, low ventilation and light, overcrowding

Aspect	Traditional rural house	Needed improvement
Walls	Mud, wattle-and-daub, stone, bamboo	Plinth protection, stabilised mud, bands
Roof	Thatch, mud terrace, country tiles	Fire-proof and leak-proof, bamboo / MCR tiles, RCC / precast
Floor	Mud / cow dung	Cement / brick floor with damp proof
Cooking	Chulha in room, smoke	Smokeless chulha with chimney, LPG, kitchen vent
Sanitation	Open defecation	Twin-pit latrine, bath area
Water	Well, pond	Hand pump / tap, RWH

2. Traditional Practice of Rural Housing

Every region has developed a house type suited to its **climate, materials and way of life**.

Region	Traditional house
Rajasthan, Gujarat (arid)	**Mud / adobe walls, flat mud roofs (terrace), circular "Bhunga" huts** of Kutch (cyclone and earthquake resistant), thick walls, small openings, lime wash
Kashmir (cold, seismic)	**Dhajji-dewari** (timber frame with brick or stone infill), sloping roofs for snow; **taq** system (timber-laced masonry)
Himalayan states (Uttarakhand, HP)	**Stone masonry with timber bands, slate roofs, Kath-kuni** (timber-stone cribbing)
North East (Assam, Manipur)	**Ikra (bamboo reed plastered with mud) walls, Assam-type houses, stilt houses (chang ghar)** against floods and animals, bamboo and thatch roofs
Kerala, coastal belts	Laterite stone and timber, **steep tiled / thatch roofs** (to drain heavy rain), wide verandas
Indo-gangetic plains (UP, Bihar, Bengal)	**Mud / mud-brick walls, thatch or country-tile (khaprail) roofs**, bamboo; plinth raised against floods; Bengal **"Bangla" (curved) roofs**
Deccan plateau and South	Stone / mud walls, **Madras terrace and tile roofs**, central courtyards
Tribal areas	Round huts of bamboo, mud, grass; shifting dwellings

- **Lessons:** use of **locally available materials**, **steep roofs in heavy rain**, **stilts in flood**, **thick walls in hot-dry**, **light and flexible structures (bamboo, timber) in earthquake**, **low height** and **compact plan** for strong winds, **courtyard** for light and ventilation

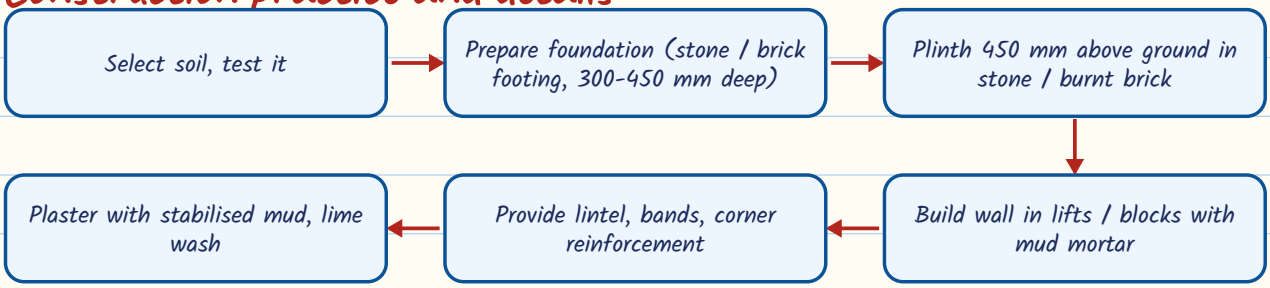
3. Mud Housing Technology

Earth (mud) is the oldest and the cheapest building material, available at site for free. Earthen buildings are **cool in summer, warm in winter**, need no fuel to make, are fully **recyclable**, and have **very low embodied energy**. Weakness: low strength and **erosion by water**.

Methods of mud construction

Method	Description
Cob (monolithic mud walls)	Wet soil mixed with **straw / husk**, built up in **layers (lifts) of 300–600 mm height**, allowed to dry (a day) before the next; wall thickness **300–600 mm**, trimmed after drying
Adobe (sun-dried mud blocks)	Soil + straw moulded in wooden moulds (e.g. **300 x 150 x 100 mm**), **sun dried for 14 days**, laid in **mud mortar**
Rammed earth (pise)	Moist soil (**about 10% water**) **compacted in layers of 100–150 mm** in wooden formwork; strong (**1.5–3 MPa**) and durable; thickness 300–450 mm
Wattle and daub (ikra, bahareque)	Bamboo / wood frame, woven mat (**wattle**), plastered with **mud and cow dung (daub)**; **light and earthquake resistant**
Compressed stabilised earth blocks (CSEB)	Soil + cement / lime pressed in a machine; stronger and uniform
Mud-brick (kachcha brick)	Moulded and sun dried, like adobe

Construction practice and details

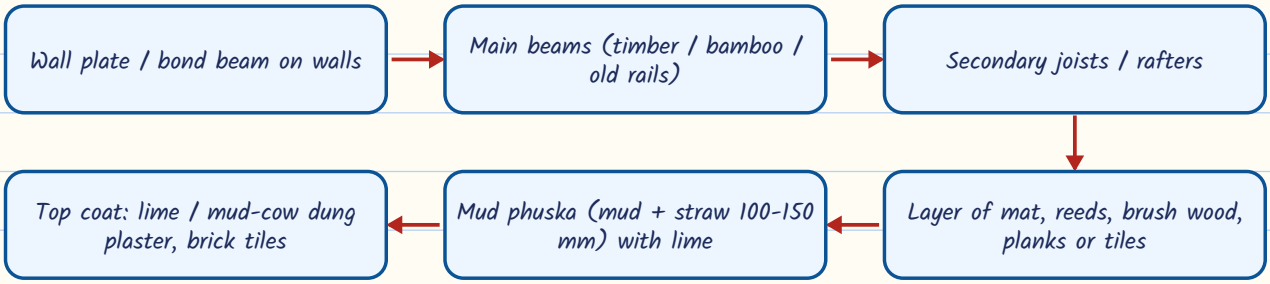


- **Foundation and plinth: stone or burnt brick footing** (not mud) at least **300–450 mm** deep and a **plinth of 450 mm** with a **damp-proof layer** (stone slab / plastic sheet) so that capillary water does not damage the mud wall
- **Wall thickness: not less than 300 mm** (cob 450–600 mm); **height max 8 times thickness**, i.e., not more than **2.7–3 m (single storey)**; **length of wall between cross walls max 6 m**; **openings** small, **not more than 1/3** of wall length, **at least 1 m from corners**
- **Lintels** of wood, bamboo or **RCC**, **bands** of timber / bamboo / RCC at **lintel and eave level** (earthquake safety); **corner reinforcement** (bamboo / steel / geogrid)
- **Protection of walls: wide roof overhang (min 750 mm)**, **plaster** (mud + straw + cow dung, **lime-cement**), **bitumen or lime wash** (coatings), **stone aprons**, **gutters and drains** away from walls, **no leaking taps and pipes**

- **Disadvantages:** erodes in rain and floods, **low strength**, termites, cracks, **needs yearly maintenance**; poor image

4. Mud Roofs

Mud roofs **(flat or low pitched terraces)** are common in **dry and semi-arid areas (Rajasthan, Gujarat, MP, Deccan)**, where rainfall is low.



- **Mud phuska roof:** soil mixed with **chopped straw / husk**, laid in 100-150 mm and well rammed, with **slope 1 in 30 to 1 in 20**; finished with **mud plaster and a layer of lime or bitumen**; **repaired every year before rains**; **weight 250-300 kg/m^2** (heavy, so **bad for earthquakes**)
- **Improved mud roof: waterproofing** layer **(polythene sheet / bitumen coated felt / lime-brick-bat layer)**, **good drainage with spouts**, **parapet 150 mm**, **stabilised mud**, **roof bands**; **thermal comfort is excellent**, because mud has a high heat capacity and low conductivity
- **Other roofs: thatch (see below)**, **country tiles (khaprail) on bamboo / timber rafters**, **Mangalore tiles**, **MCR tiles**, **GI / AC / bamboo mat sheets**, **Madras terrace**, **brick-on-edge arches**, **precast plank-and-joist**, **filler slab**

Design points for pitched roofs in rural areas

* **Slope 30 - 45 degrees** for tiles, **45 - 60 degrees** for thatch (rapid water shed)
* **Overhang (eaves) 600-900 mm**, rafters tied; **hip roofs** better in cyclone areas than gable roofs
* Light roof (tiles / thatch / sheets) for **earthquake zones**; **tie roof to wall** firmly **(anchor bolts, wire ties)**

5. Characteristics of Mud (Soil as a Building Material)

Soil is a mixture of **gravel (above 2 mm)**, **sand (0.06–2 mm)**, **silt (0.002–0.06 mm)** **and clay (below 0.002 mm)**. Clay is the **binder**, sand/gravel the **filler / skeleton**.

Property	Desirable value / meaning
Composition for mud walls	**Clay 15 – 25%, silt 10 – 30%, sand 50 – 70%, no organic matter**
Compressive strength	**0.5 – 3 MPa** (dry), depends on clay, compaction and moisture; **wet strength much lower**
Plasticity	**Liquid limit 25 – 50%, plasticity index 7 – 18** - too much clay cracks, too little crumbles
Shrinkage	**Clay shrinks 5–10% on drying: cracks**; reduce by adding **sand / fibre**
Density and thermal property	**1600 – 2000 kg/m^3**; thermal conductivity **0.5 – 0.9 W/mK** (good insulation), high heat capacity (**thermal lag**)
Water absorption and erosion	High: **must be protected**; swelling in clay
Fire and sound	Fireproof; good sound insulation

Field tests of soil

Test	Procedure	Meaning
Ribbon (roll) test	Roll a moist ball into a thread of 3 mm dia; 150–250 mm length without breaking	Good clay content
Dry strength / cigar test	Make a ball of 20 mm, dry; try to crush	Resists = good binder
Jar (sedimentation) test	Shake soil with water in a transparent jar; read layers after 1 and 24 h	Percentage of sand, silt, clay
Shrinkage box test	Fill a **60 x 60 x 600 mm** box with moist soil, dry, measure shrinkage	Cracks and length decrease should be below 15–20 mm
Drop (ball) test	Drop a 40 mm ball from 1.5 m	Breaks into few pieces = OK
Smell and visual test	Musty smell shows organic matter, which is bad	Reject soil with humus

6. Soil Stabilization

Soil stabilisation improves the **strength, water resistance (durability)** and reduces **shrinkage and swelling** of the soil by mechanical, chemical or physical means.

Stabiliser	Dosage	Best for / effect
Compaction (mechanical)	Rammed layers, compaction **at optimum moisture**	Cheap, increases density and strength
Cement	**5 – 10%** by weight (**sandy soils, PI below 15**)	Hydration gives bonding, high strength (3-5 MPa), reduces swelling; cure for 7-28 days
Lime	**6 – 12%** (**clayey soils, PI above 15**)	Reacts with clay (pozzolanic): **strength, less plasticity**; slow
Bitumen / cut-back / emulsion	**4 – 6%** (sandy, low clay)	**Water repellent** (waterproofing); no strength gain
Fly ash, rice husk ash, slag (with lime)	10-30%	Pozzolanic, cheap
Fibres: straw, jute, coir, sisal, bamboo chips	**0.5 – 2% by weight**	**Reduce cracking, improve tensile strength**
Cow dung, molasses, gum, tannin	Plaster and wash coats	Traditional, water resistant
Gypsum, calcium chloride, enzymes, polymers, resins	Small %	Special / modern chemical stabilisers

Rules of thumb

* **Sandy soils**: use cement or bitumen; **clayey soils**: use lime (or lime + cement), fibres
* **Cement-soil blocks**: soil (sieved) + 6-8% cement, water to optimum moisture content, **press, cure for 14 days under shade** (keep damp)
* **Stabilised soil must be tested**: wet compressive strength not less than **1.5-2 MPa**; water absorption max 15-20%

Example: mix for 1000 stabilised blocks

Block size 230 x 190 x 100 mm = 0.00437 m^3; compacted density 1900 kg/m^3 => 8.3 kg per block => 8300 kg soil mix per 1000 blocks.

With 7% cement: cement = 0.07/1.07 x 8300 = **543 kg (11 bags)** and dry soil = 7757 kg

Cost advantage = about 30% lower than burnt bricks (no fuel).

7. Fire Treatment for Thatch Roof

Thatch (paddy straw, wheat straw, grass, reed, palm leaves, coconut leaves) is cheap, light and an excellent **insulator**, but is very **flammable** and is attacked by rain, insects and fungus; **life of 5-10 years**. **Fire** is the biggest danger in villages (cooking fires, sparks, short circuits).

Fire retardant treatments

Treatment	Method
Mud-lime or mud-cement slurry coating	Spray / apply a plaster of **mud + cow dung + lime / cement** on the thatch surface (5-10 mm)
Chemical solutions (retardants)	**Borax + boric acid** (ratio 1:1), **ammonium sulphate / phosphate, alum, sodium silicate, zinc chloride**: a solution of about **10-15%** is sprayed or the thatch is **dipped** and dried
Lime wash / Cement-lime paint	Over the top
Thatch over fire-proof underlayer	Lay **bamboo mat plastered with mud**, **or a tarred / GI sheet / bricks** beneath
Use of fire barrier	Keep thatch house **at least 5-10 m apart** and **chulha away** from thatch, **chimney with spark arrestor**, **separate kitchen** with **fire resistant roof** (tiles, sheets)

- **Other treatments: neem oil or copper sulphate for fungus and insects**; **bitumen emulsion for waterproofing**; **steep slope of 45-60 degrees**, thickness **150-250 mm**, **tied** to bamboo battens by coir / wire; **replace damaged thatch** before monsoon
- **Fire safety practices: water pots / pits in the village**, community firefighting awareness, **electrical wiring with proper insulation**, **smokeless chulha with chimney**, **avoid open flame lamps near thatch**

8. Rural Housing Programmes

Programme	Details
Indira Awaas Yojana (IAY, 1985-2016)	Grants for **BPL families (SC/ST, freed bonded labour priority)**, 40% quota to SC/ST; started as part of RLEGP / JRY
Pradhan Mantri Awaas Yojana - Gramin (PMAY-G, 2016)	**Housing for All in rural areas**; target **2.95 crore houses**; beneficiaries from **SECC 2011** data (kutcha and houseless); **assistance Rs 1.20 lakh (plains) and Rs 1.30 lakh (hills, NE, difficult areas)**; **minimum house size 25 sq.m**, with a **cooking space**; **toilet** via **SBM-G (Rs 12,000)** and **90-95 days MGNREGA wages**; **loan up to Rs 70,000** optional; **payment through DBT**; **Geo-tagging (AwaasApp)**, **Rural Mason Training**
Other state schemes	Samagra Awaas, Mukhyamantri Gramin Awas, Biju Pucca Ghar, Indiramma houses (AP)
Rural Housing Fund (NHB), HUDCO, NABARD	Credit for rural housing
Rural Building Centres, Nirmithi Kendras	Technology and training
Convergence	Jal Jeevan Mission, Ujjwala, Saubhagya, **MGNREGS**, PMGSY

- **Objectives:** remove **kutcha houses**, ensure **safe, durable, hygienic** houses with **basic services**; **empower the women (houses in the name of women / jointly)**; **local typology designs** (regionally suitable) given to beneficiaries; **disaster resistant features** are encouraged
- **Implementation: gram sabha** selects beneficiaries, funds in **instalments (e.g., 25%, 60%, 15%)** linked to **plinth, lintel, completion** stages, with **photo geo-tagging**; **technical support from master masons, rural mason training (RMT)** - community and **NGOs**
- **Issues:** delays, low unit assistance versus the cost of material, **rising cost of materials**, lack of technical guidance, poor quality, **land titles for landless**, **non-engineered building**, difficult hilly regions; **solutions: cost-effective technology (rat-trap, stabilised blocks, filler slab, MCR tiles), housing finance, training, monitoring, convergence**

Typical low cost rural house design (about 25-30 sq.m)

* Plan: **one room (3 x 3 m) + kitchen + verandah + toilet**, extendable by adding rooms **(core house concept)**

* Foundation: **stone or brick masonry in 1:8 mortar**; **plinth 450 mm**; walls: **stabilised mud blocks or fly ash bricks in 1:6 mortar (rat-trap bond), 230 mm**
* Roof: **MCR tiles / GI sheets on bamboo or steel trusses, or filler slab / precast plank-joist**, with **0.75 m overhang**
* **Smokeless chulha**, **twin-pit latrine**, rain water harvesting, **cost about Rs 1.2 – 2.5 lakh** (varies)

Design considerations for rural houses

- **Climate-responsive** orientation (long axis east-west so that the long walls face north and south; small openings on the west), **cross-ventilation**, **shading**, **thermal comfort**
- **Disaster-resistance** (flood, cyclone, earthquake, fire) as per **IS 4326, IS 13827 (earthen buildings in seismic zones), IS 13828 (low-strength masonry), IS 15883 (guidelines for construction)**
- **Social needs: cattle shed separated**, **grain store**, **courtyard**, **women's working space**, **toilet attached**

9. Bamboo in Rural Housing

- Bamboo is abundant in **NE India, Kerala, Bengal, Odisha**; grows to maturity in **3–5 years**; **tensile strength about 100–300 MPa**; light and flexible, so **good for earthquakes**
- **Uses: posts, rafters, trusses, wall frames (wattle), mats (bamboo mat board, ikra), bamboo reinforced concrete, scaffolding, floor, doors / windows**
- **Treatment is a must: borax-boric acid (1:1.5), 5% solution by the "Boucherie / soak and diffusion" method for 7–10 days**, or **smoking**, or **chemical-pressure treatment**; protect from **moisture** with a raised footing on a stone or concrete base (never bury bamboo in the ground)
- **Joints** are the weak point: use **bolts, lashing with GI wire / coir rope, steel brackets, and mortar filling inside bamboo node**
- **Bamboo mat corrugated sheets (BMCS)** are a cheap roofing; **Bamboo mat board** in partitions

10. Flood, Cyclone and Fire Safe Features in Rural Houses

Hazard	Features
Flood	**Plinth above high flood level (HFL)**, stilt / platform house, **stabilised mud or burnt brick plinth**, water-resistant plaster, **raised platform (community)**, **no mud walls below flood level**
Cyclone / strong wind	**Hip roof, slope 22–30 degrees**, low height (single storey), **anchor roof to wall by ties and bolts**, **small overhang**, strong foundation, **no heavy tiles unattached**, shelters
Fire	**Fire-retarded thatch, separate kitchen, brick / stone chimney, spacing 5–10 m**
Earthquake	**Light roof, symmetrical plan, bands, corner reinforcement**, no heavy mud roofs
Termite and rain	**Termite barrier (soil treatment), wide eaves, drainage around the house, lime wash**

11. Annual Maintenance Calendar of a Mud House

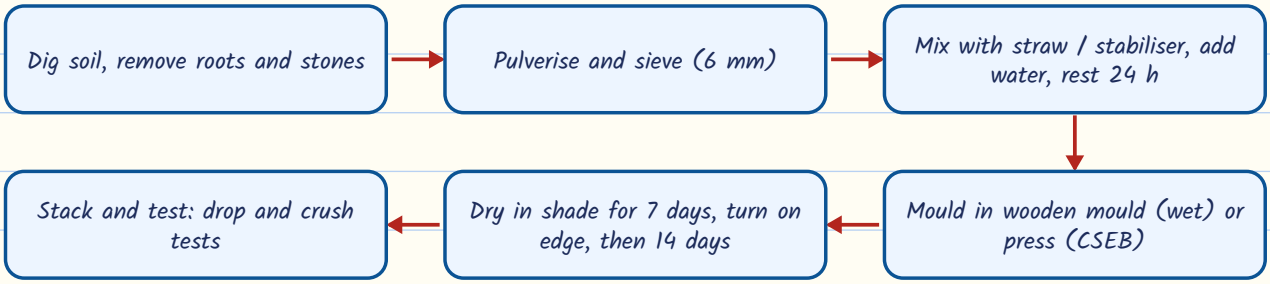
Time	Work
Before monsoon (April–May)	Replaster walls with mud-cow dung (or stabilised mud); repair cracks; repair roof (phuska / thatch / tiles); clear drains and gutters
During monsoon	Inspect leaks, keep surroundings dry, protect plinth, remove vegetation near walls
After monsoon (Oct–Nov)	**Lime wash** with alum / salt; fill cracks with mud slurry; termite check
Every 3–5 years	Re-lay thatch; replace rotten timber / bamboo; strengthen foundation aprons
Every 10 years	Replace damaged mud blocks, repair the plinth, re-do major finishing

12. Cost Estimate of a Core Rural House (Illustrative)

Item	Approximate share of cost
Foundation and plinth	15%
Walls (stabilised mud / fly ash bricks)	30%
Roof (MCR tiles / sheets / precast)	25%
Doors, windows, flooring	12%
Plaster, painting, finishing	8%
Toilet, water, electricity	10%

Total for a 25 sq.m house is about **Rs 1.5 – 2.5 lakh** (as per local rates); **PMAY-G assistance (Rs 1.2 / 1.3 lakh)** + **MGNREGS wages** + beneficiary contribution (own labour) fill the cost gap. Use of **local labour and materials** cuts it by 20-30%.

13. Production of Adobe and Stabilised Blocks (Step by Step)



- **Quality checks:** no cracks, sharp edges, **dry compressive strength min 2.5-3.5 MPa**, wet strength min 1.5 MPa after 24 h soaking, and **erosion test (spray of water 1 h under 0.5 m head)** with depth of pitting less than 15 mm
- **Mud mortar:** same soil as block, with **more sand and fibre**, thin joints (10-12 mm), **fill all joints**; **cement-mud mortar 1:12** for improved strength
- **Quantity:** a 3 m x 3 m room with 300 mm walls (2.7 m high) needs about **1100-1300 adobe blocks** (300 x 150 x 100 mm)

14. Rural Building Centres and Technology Transfer

- **Rural Building Centres (RBCs)** and **Nirmithi Kendras** (Kerala), **HUDCO building centres**, **KVKs**, NGOs like **Auroville Earth Institute, Hunnarshala (Gujarat), Laurie Baker Centre (COSTFORD)** train masons, show **demonstration houses**, and **produce stabilised blocks, tiles, door and window frames**
- **Rural Mason Training (RMT)** under PMAY-G certifies masons for **disaster-resistant and cost-effective construction**
- **Role of institutions: CBRI Roorkee** (mud and low cost technologies), **SERC Chennai**, **BMTPC** (guidelines and Vulnerability Atlas), **NIRD & PR Hyderabad** (training for rural development), **IITs and NITs**
- **Participatory approach:** involve **women, SHGs and gram panchayat**; **owner-driven reconstruction** (post-disaster Gujarat, Maharashtra) gave good results

Remember for exam

* Mud house types: cob, adobe, rammed earth, wattle-and-daub; wall thickness not less than 300 mm; plinth 450 mm
* Good mud soil: sand 50-70%, silt 10-30%, clay 15-25%; tests: ribbon, jar, shrinkage box, drop test
* Stabilisers: cement 5-10% for sandy soil, lime 6-12% for clayey soil, bitumen 4-6% for waterproofing, fibres 0.5-2%
* Thatch fire retardants: borax-boric acid, ammonium sulphate / phosphate, alum, mud-lime coating
* PMAY-G: 2.95 crore houses, Rs 1.2 lakh (plains) / 1.3 lakh (hills), 25 sq.m with cooking space, SECC data, geo-tagging

Important questions

1. Describe the traditional practices of rural housing in different regions of India.
2. Explain the mud housing technology: cob, adobe, rammed earth and wattle and daub.
3. Discuss the construction of mud roofs and their maintenance.
4. Explain the characteristics of mud. How is soil tested for use in mud construction?
5. What is soil stabilisation? Explain the methods with the stabilisers and their dosage.
6. Explain the fire treatment for thatch roofs.
7. Describe the rural housing programmes of the government (IAY / PMAY-G).

8. Design a low cost rural house with suitable materials and technology.